Q10 Count Div

```
CREATE a variable count = 0
LOOP for all numbers between A and B
    IF the number if divisible by K
        ADD 1 to count
RETURN count                          too slow
```

```
IF A if divisible by K
    start at A
ELSE
    start at the first number after A that is divisible by K
RETURN (B - start) // K + 1
```

Q9 Missing Integer

```
CREATE a set for buffer
LOOP for all numbers in A
    IF the number is positive
        ADD the number to buffer list
SORT the buffer
CREATE a variable last = 0
LOOP for the buffer list
    If the number <= last + 1
        SET last to the number
    ELSE
        BREAK the loop
RETURN last + 1
```

Q8 Passing Cars

```
CREATE a list holding count of cars going west
CREATE a variable count
LOOP for all cars
    IF the car is going west
        ADD 1 to count
    ELSE
        ADD count to the list counting west cars
        SET count to 0
IF count is not 0
    ADD count to the list counting west cars
    SET count to 0

CREATE a variable total
LOOP for all west car counts
    ADD the west cars count x count to total
    ADD 1 to count
RETURN total if total is less or equal to 1,000,000,000 else -1
```

Q7 Frog River One

```
CREATE a set with all leaves need to fall
LOOP for all falls with a index
    IF the leaf still need to fall
        REMOVE it from the set
        IF the set is empty
            RETURN the index
RETURN -1
```

Q6 Perm Check

```
SORT the list A
CREATE a variable last = 0
LOOP for sorted A
    IF this number is not equal to last + 1
        RETURN 0
    SET last = n
RETURN 1
```